Low-depth amplitude estimation via statistical eigengap estimation

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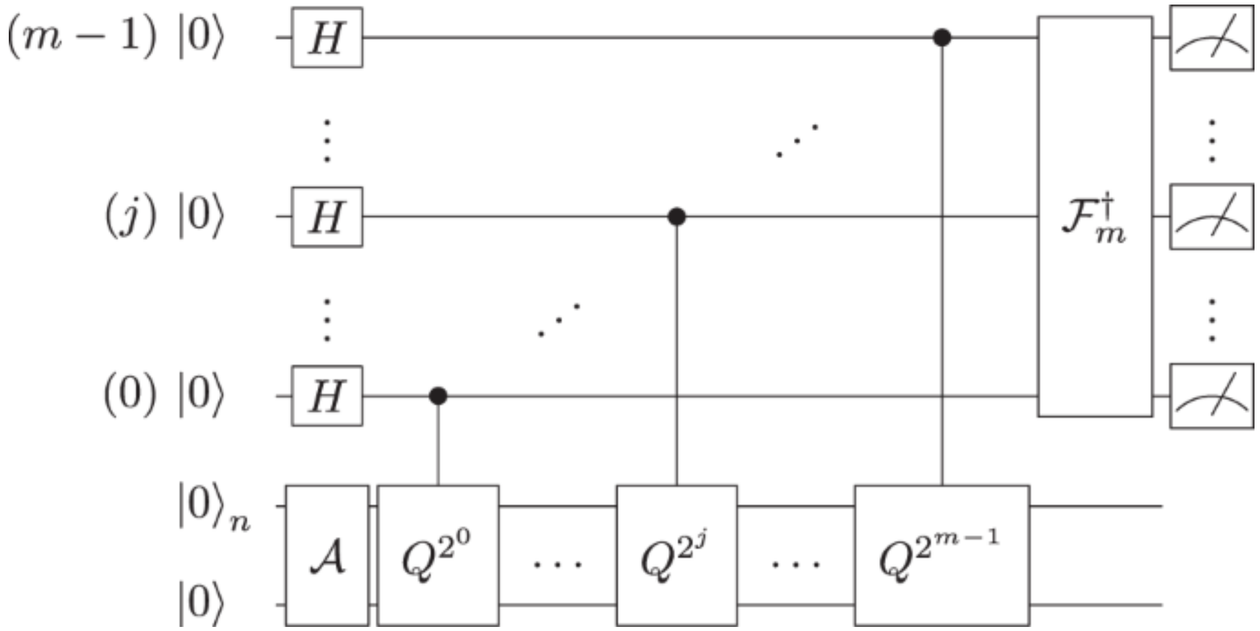
arXiv: 2603.05475

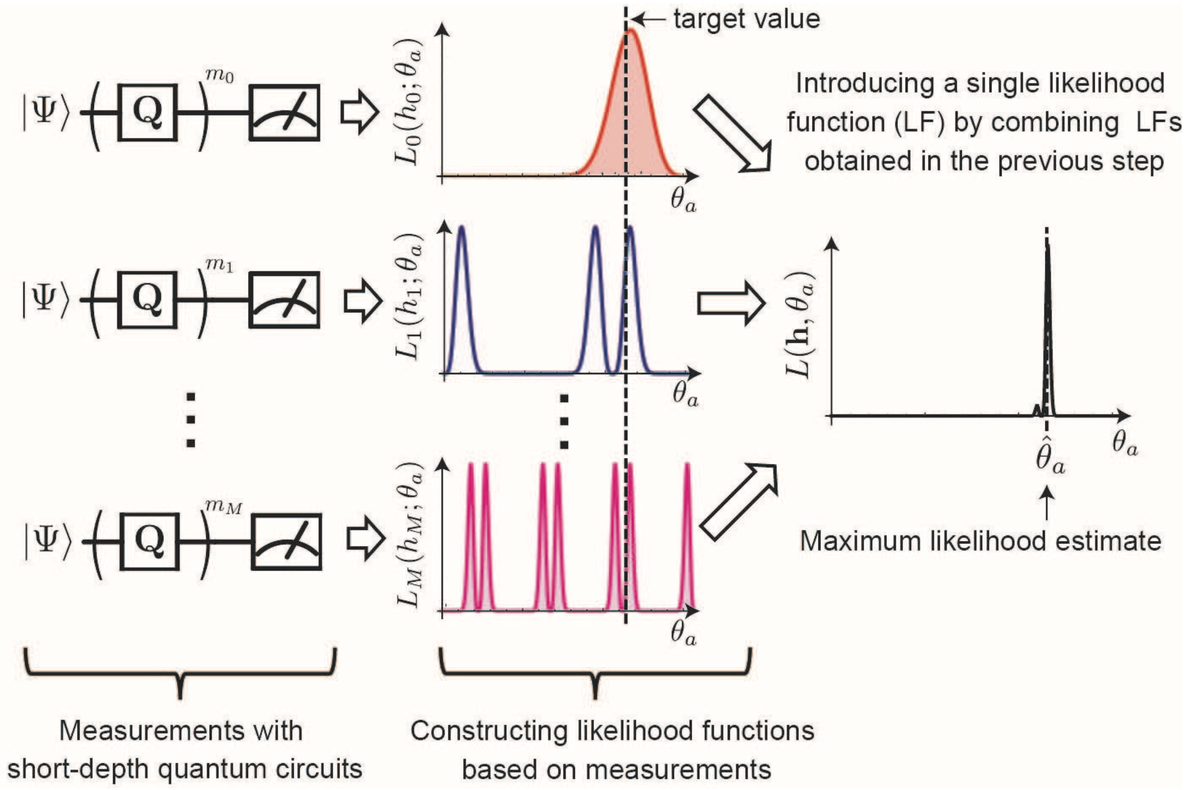
University of Oxford

Quantum Motion

QAE does not need QPE

- Modern QAE protocols only require amplitude amplification runs without QPE.





Amplitude amplification as discrete-time evolution

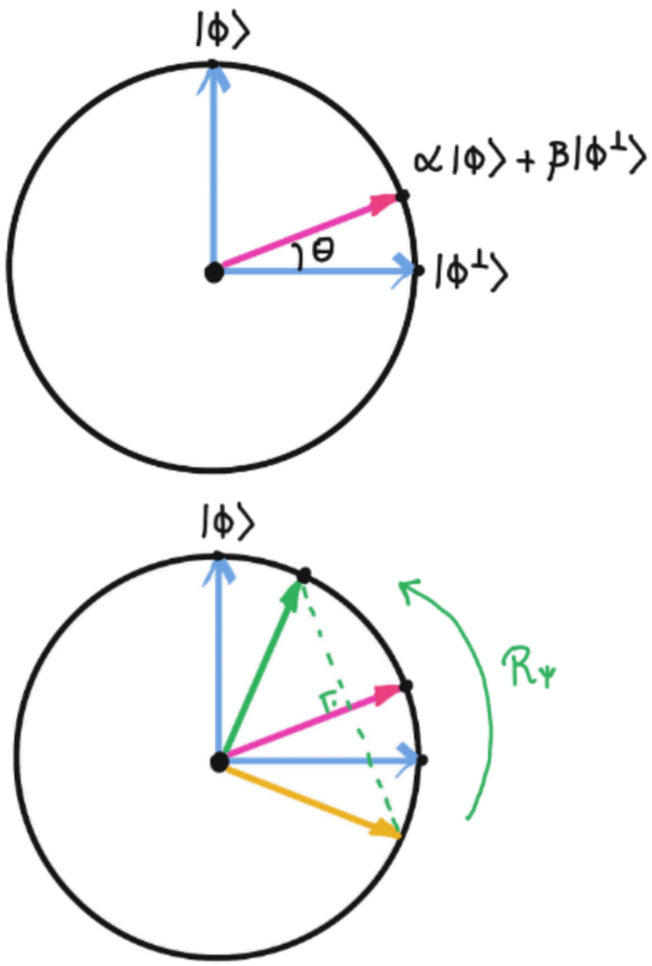
Proposition 1 (Effective Hamiltonian of amplitude amplification). *We are given Grover's walk operator* $Q = -(\mathbb{I} - 2|\psi\rangle\langle\psi|)(\mathbb{I} - 2P)$ *for some arbitrary input state* $|\psi\rangle$ *and a projector* P *such that* $a = \sin^2(\lambda) = \|P|\psi\rangle\|^2$. *Then* Q *is equivalent to a discrete-time evolution under an effective Hamiltonian* H_{eff} *where*

$$Q = e^{-iH_{\text{eff}}} \quad \textit{with} \quad H_{\text{eff}} = -2\lambda Y_\psi + \pi(\mathbb{I} - P - |\psi_b\rangle\langle\psi_b|).$$

In the 2-dimensional subspace $\mathcal{H}_\psi = \text{span}\{|\psi_g\rangle, |\psi_b\rangle\}$, *the operator* $Y_\psi = i|\psi_b\rangle\langle\psi_g| - i|\psi_g\rangle\langle\psi_b|$ *is an effective Pauli Y matrix, and therefore* Q *acts as a discrete Pauli Y rotation. In the subspace* $\mathcal{H}_\psi$, *the effective Hamiltonian has two eigenvalues which are related to the amplitude to be estimated as*

$$\pm E_{\text{eff}} = \pm 2\lambda = \pm 2\arcsin(\sqrt{a}) = \pm\arccos(1 - 2a),$$

where the last term uses the same convention as qubitization.



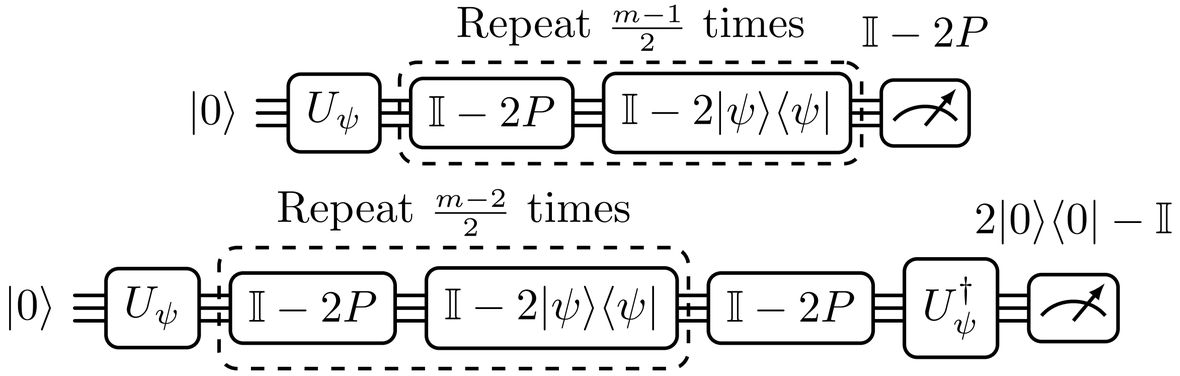
Amplitude estimation as eigengap estimation

Proposition 2 (Effective Hamiltonian eigengap estimation). *Given the discrete time evolution operator $U_t := Q^t$ under the effective Hamiltonian H_{eff} from Proposition 1, the expected value of the Pauli-like observable $O = \mathbb{I} - 2P$ yields the time-dependent expected value in Eq. (2)*

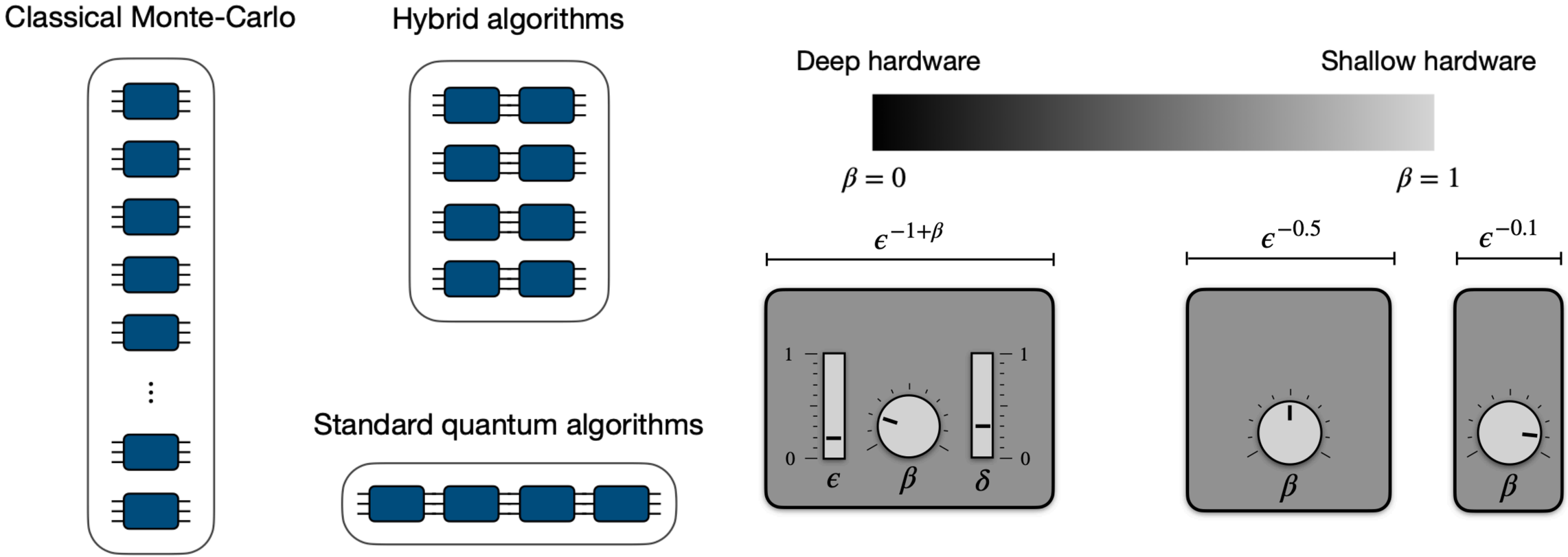
$$\langle\psi|U_t^\dagger O U_t|\psi\rangle = \sum_{E_j, E_k \in \{\pm E_{\text{eff}}\}} q_{j,k} e^{-it(E_j - E_k)}$$
$$= \frac{e^{i2\lambda}}{2} e^{it\Delta_{\text{eff}}} + \frac{e^{-i2\lambda}}{2} e^{-it\Delta_{\text{eff}}}$$

which depends on the effective Hamiltonian's energy gap $\Delta_{\text{eff}} = 2E_{\text{eff}} = 4\lambda = 4\arcsin(\sqrt{a})$. Alternatively, using the observable $O' = 2|\psi\rangle\langle\psi| - \mathbb{I}$, we obtain the time series

$$\langle\psi|U_t^\dagger O' U_t|\psi\rangle = \frac{1}{2} e^{it\Delta_{\text{eff}}} + \frac{1}{2} e^{-it\Delta_{\text{eff}}}.$$

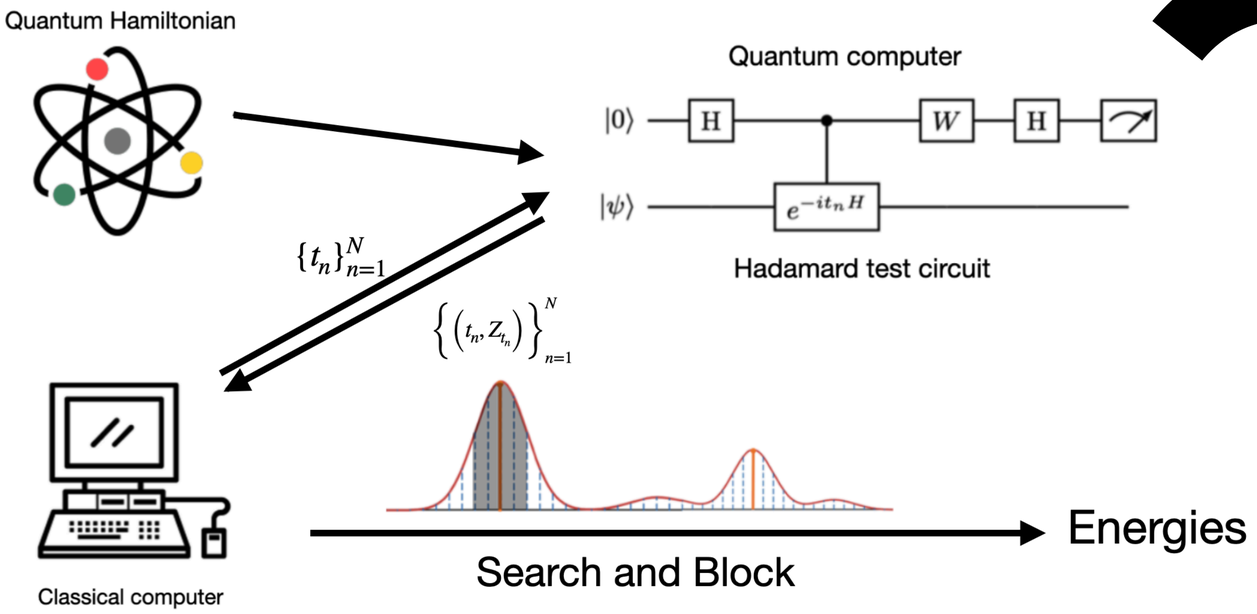


Low-depth amplitude estimation

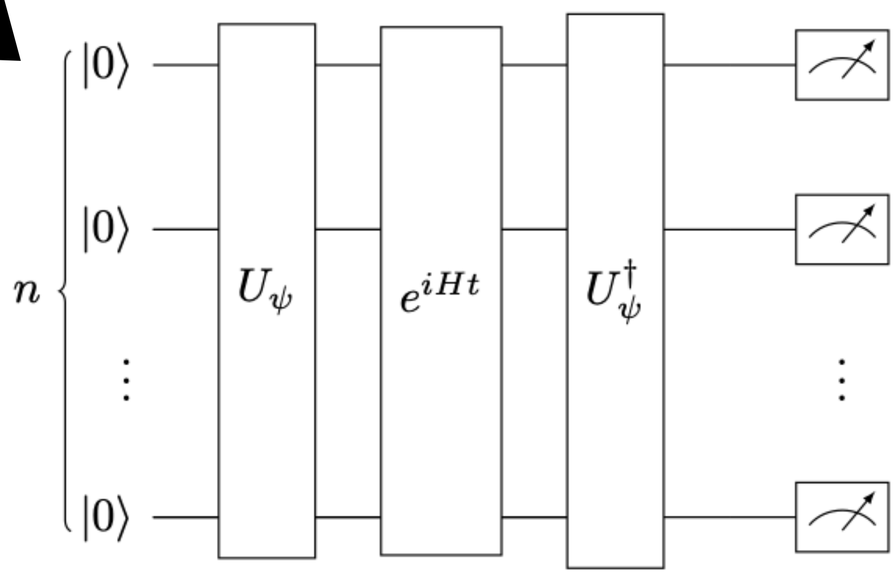


From statistical phase estimation to QAE

Replace Hadamard Test



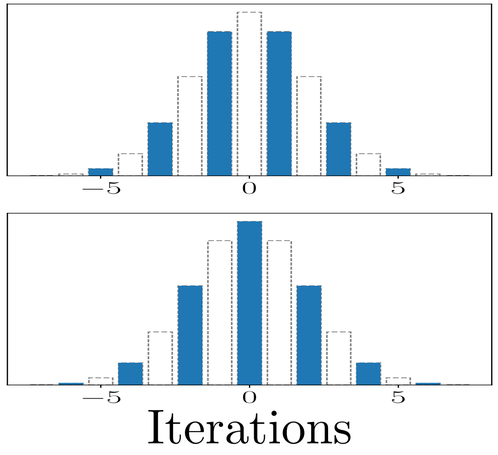
Statistical Phase Estimation



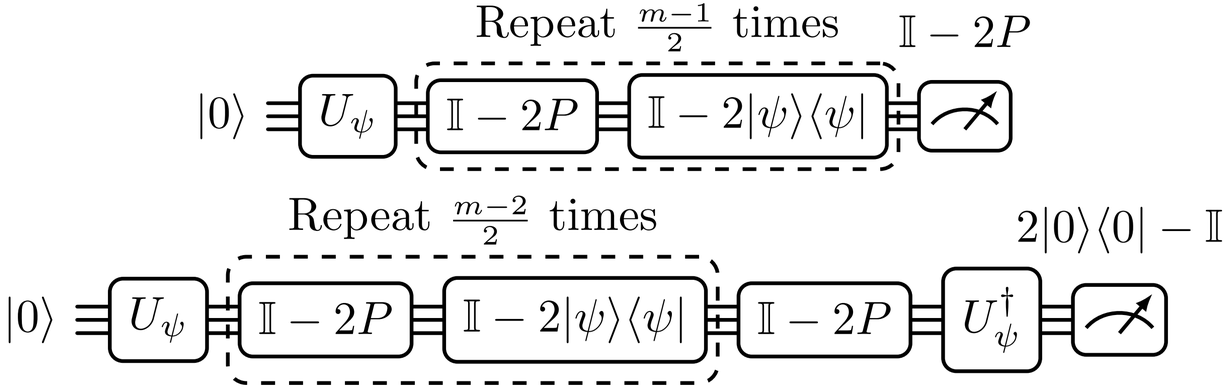
Eigengap Estimation

Amplitude Estimation

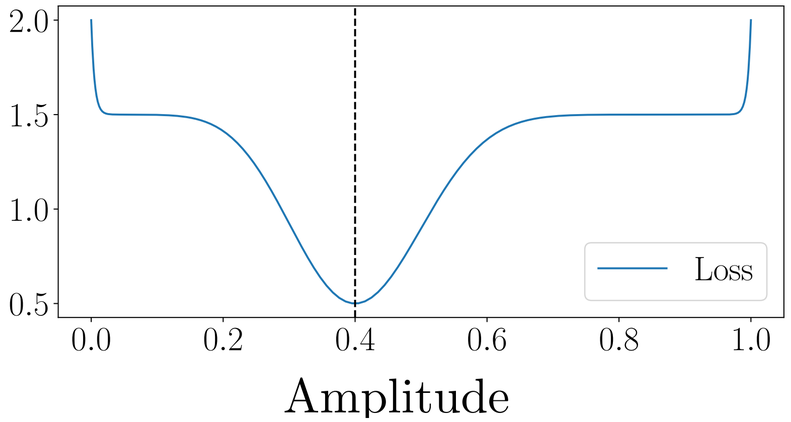
Gaussian Least Squares Amplitude Estimation



$\{m\}_{n\in[N]}$

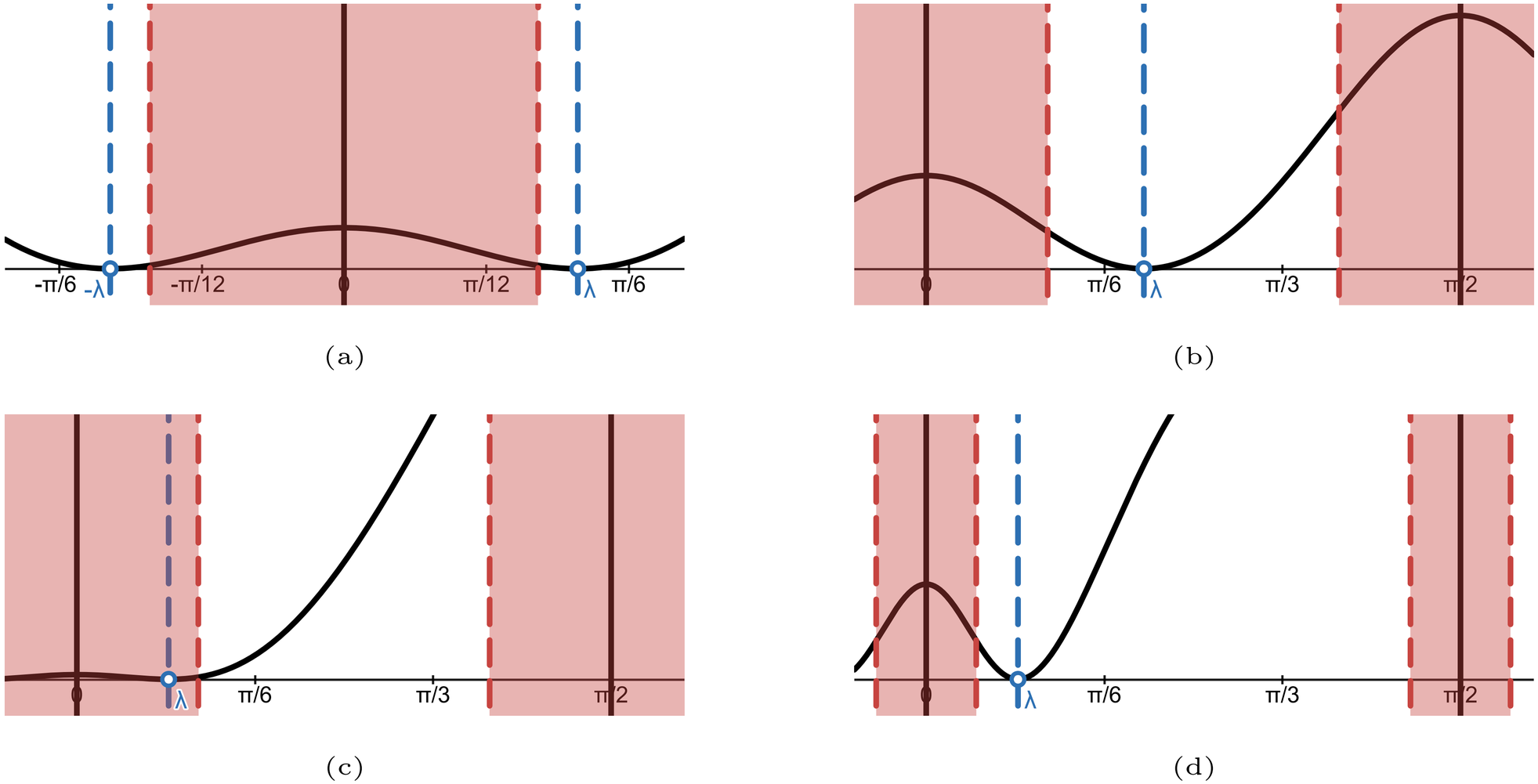


$\{(m, Z_m)\}_{n\in[N]}$

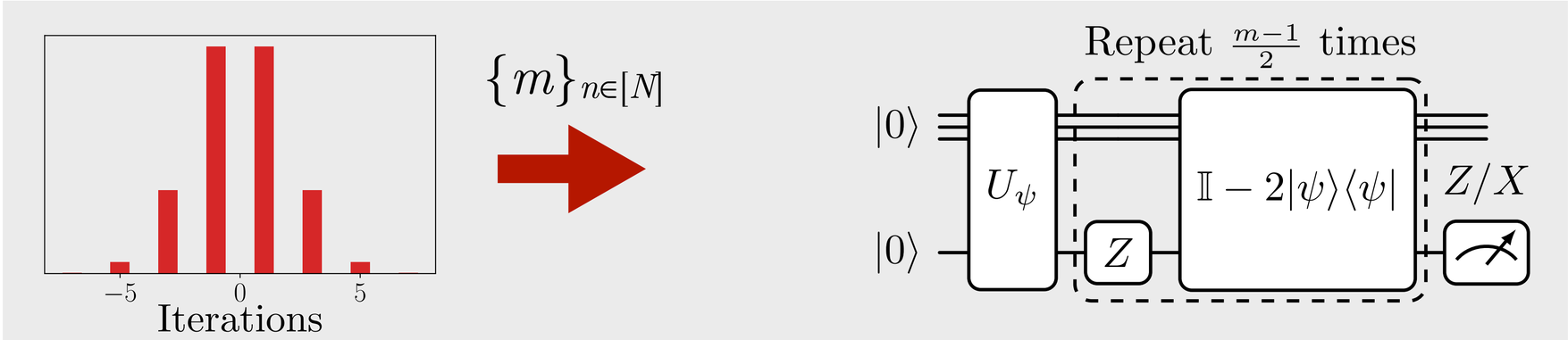


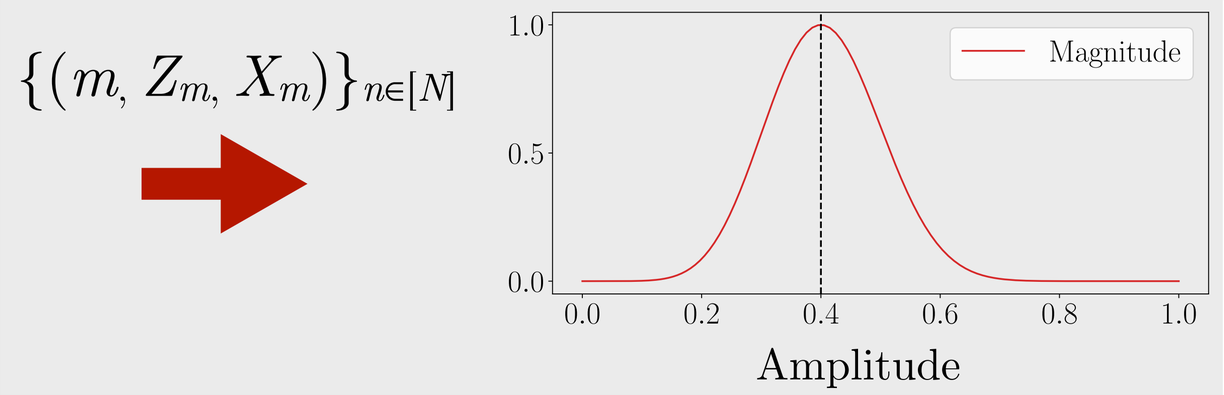
$$\tilde{\mathcal{L}}(\theta) = \frac{1}{N} \sum_{m\sim\tilde{p}_T} \left(Z_m - \cos(2\theta m)\right)^2,$$

GLSAE fails at low depths — Only cosine signals



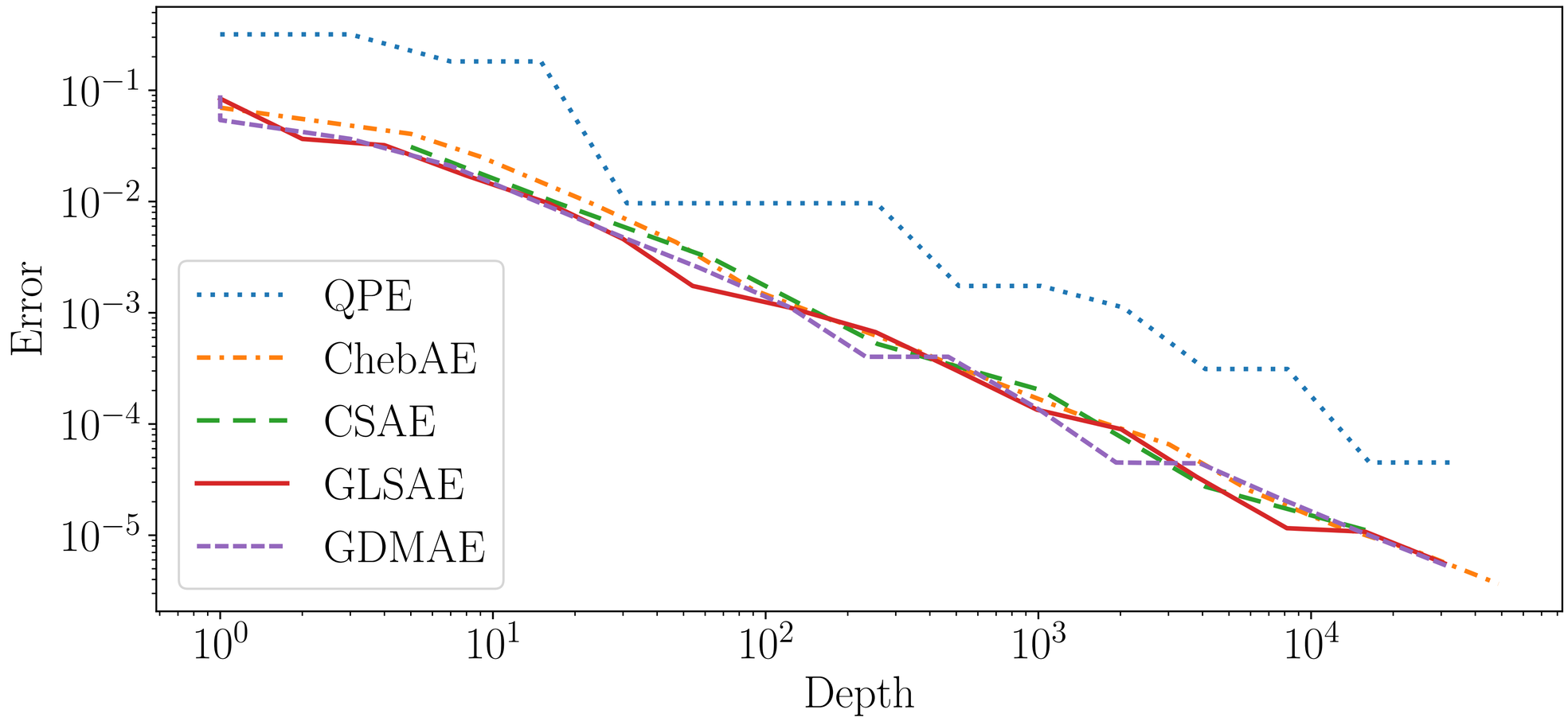
Gaussian Dual Measurement Amplitude Estimation



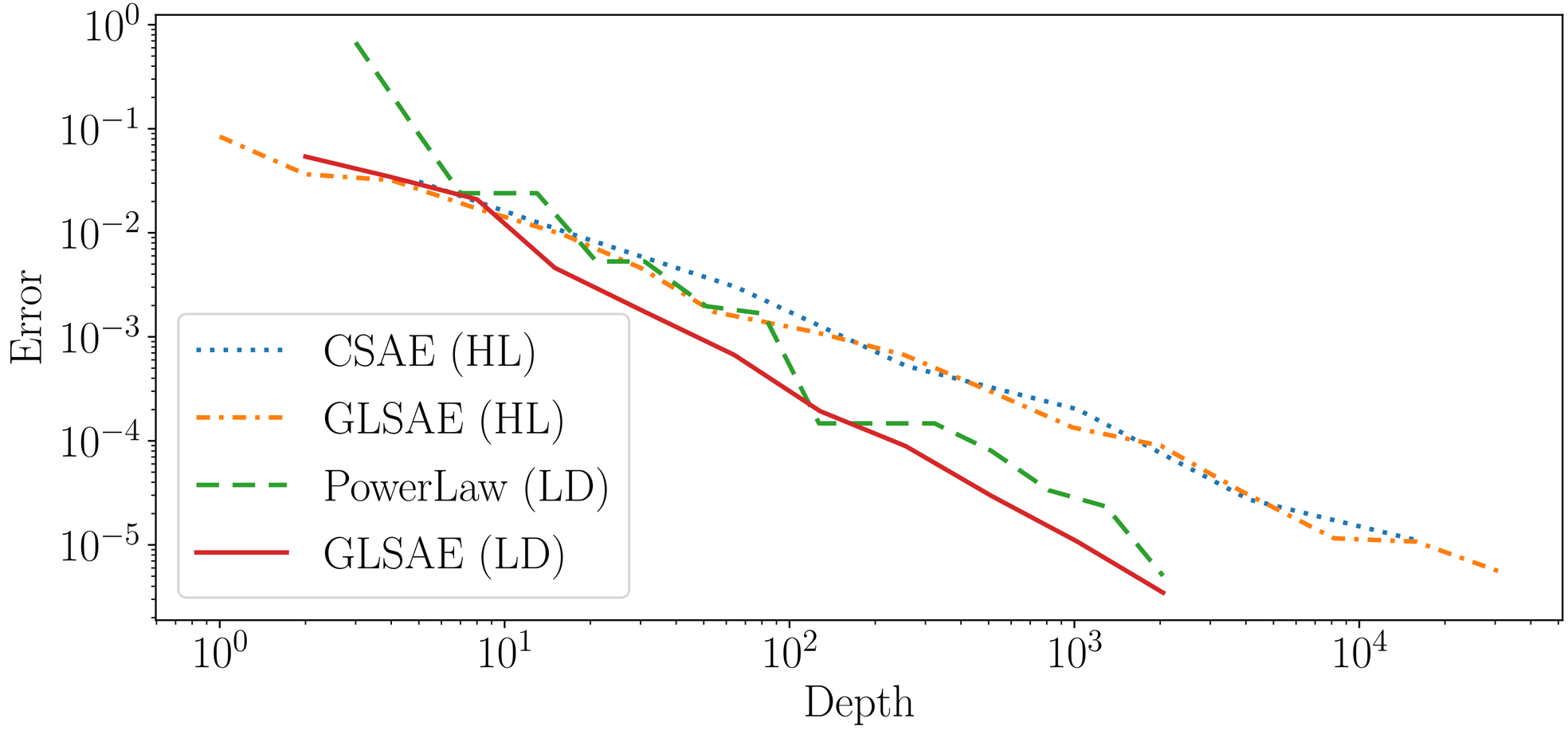


$$\tilde{\mathcal{M}}(\theta) = \frac{1}{N} \sum_{m\sim\tilde{q}_T} Z_m \cos(2\theta m) + X_m \sin(2\theta m),$$

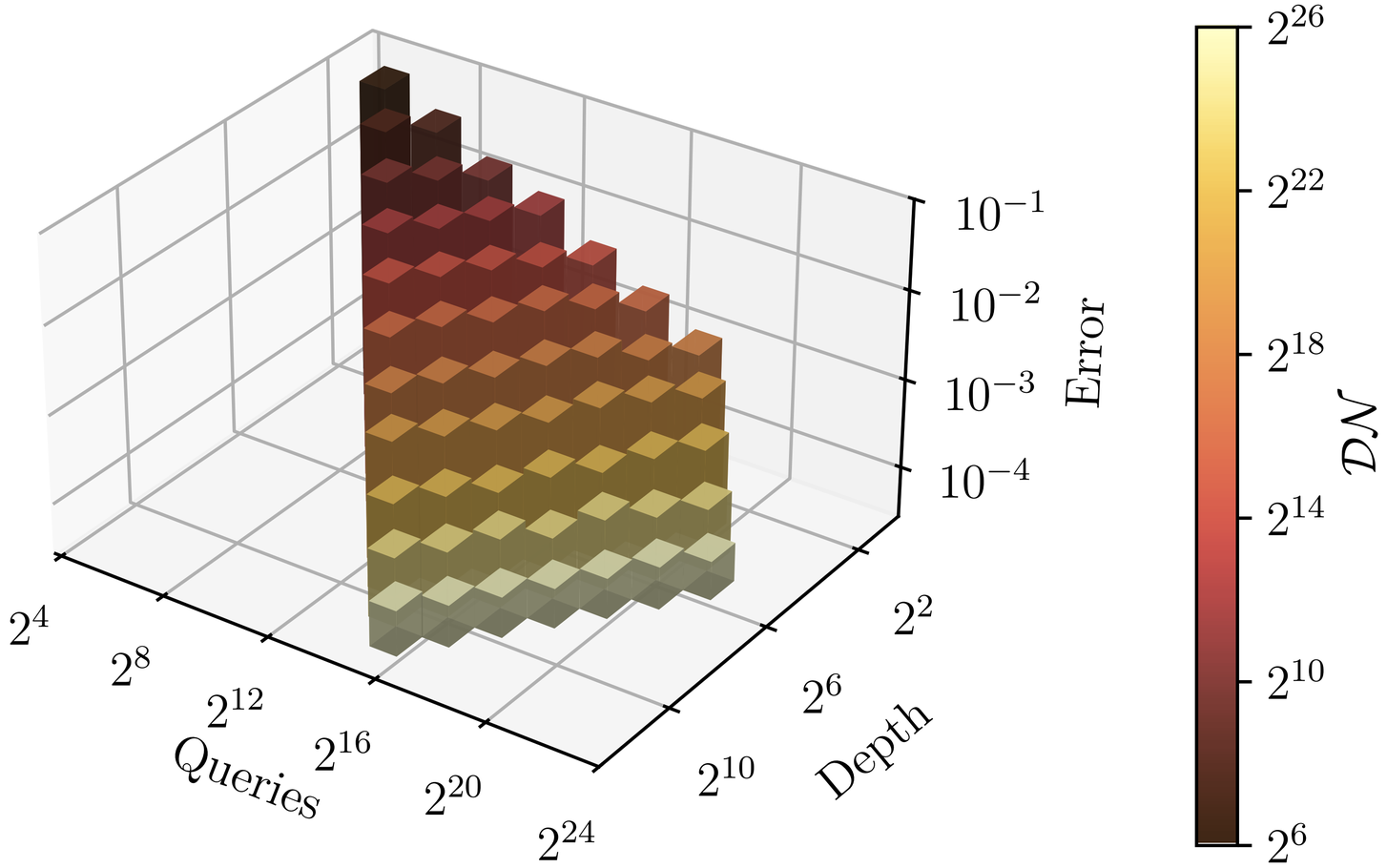
Numerical results — Benchmarking



Numerical results — Benchmarking II

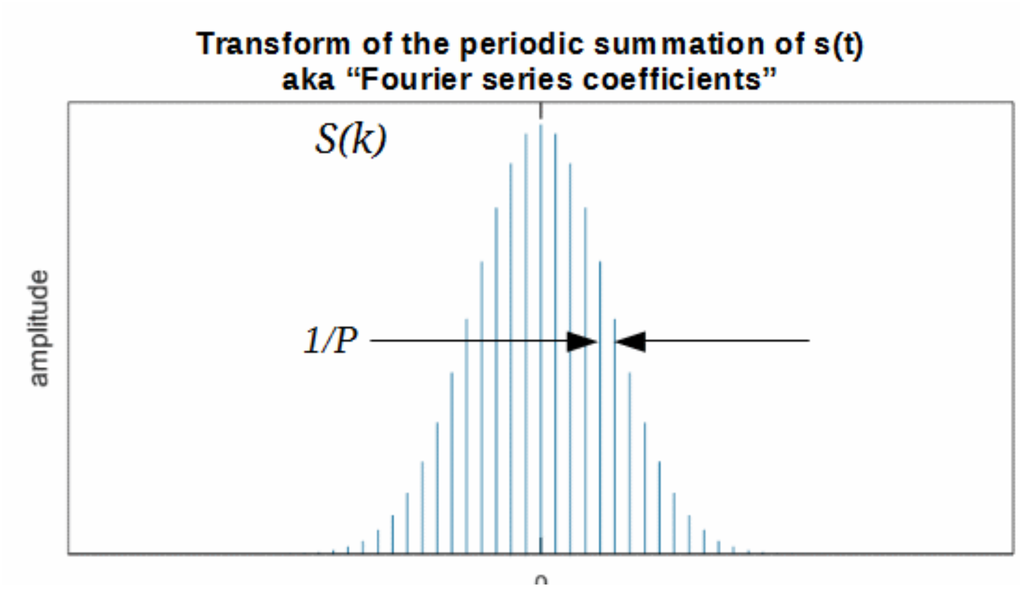


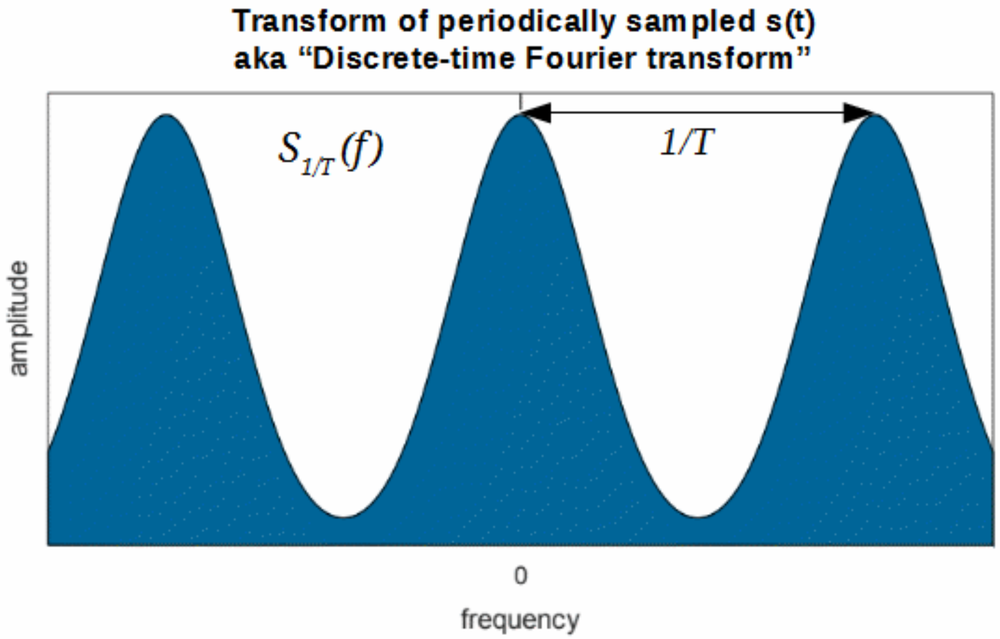
Numerical results — Depth-query invariance

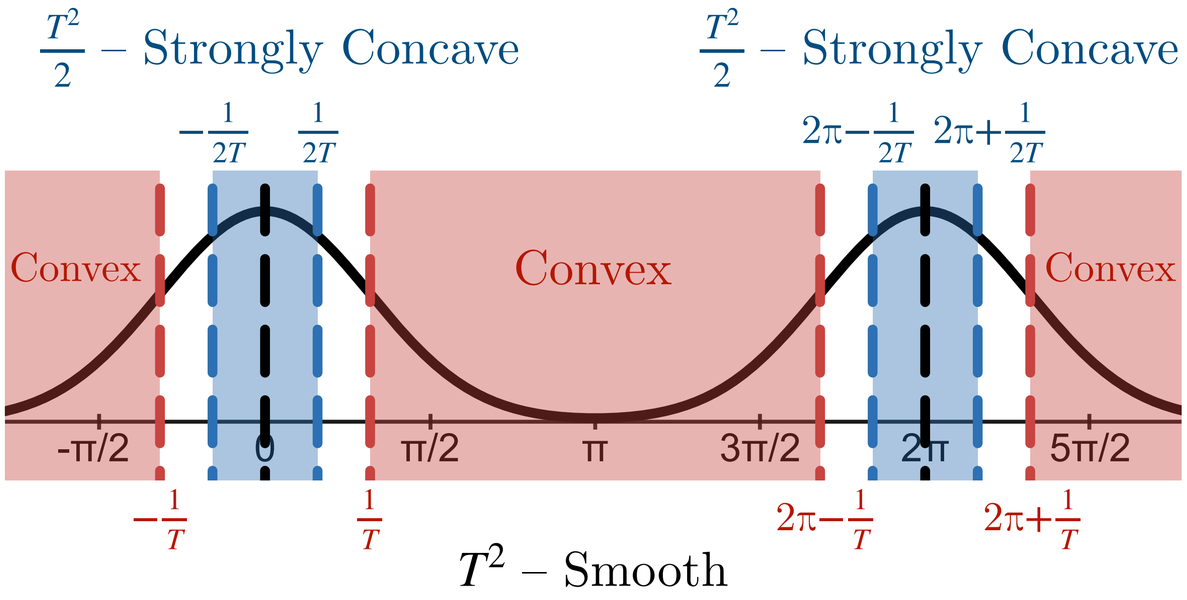


Proof ideas and strategy I

- Discrete Gaussian sample distribution + cosine signals
  - Discrete-time cosine transform of discrete Gaussian is a periodic Gaussian





Proof ideas and strategy II

- Lower quadratic bounds via strong convexity/concavity
- Upper quadratic bounds via smoothness
- Complete quadratic inequality bounds by including sampling error via Hoeffding bound and truncation errors to recover query-depth invariance